Rajagopal Somaskandan

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Summary

M.Sc. Computer Science student (graduating 2027) with hands-on experience in AI/ML engineering, Generative AI, backend development, and data engineering. Skilled in Python, SQL, FastAPI, RAG, NLP, and machine learning. IEEE-published researcher who has built supplier-ranking, fraud-detection, conversational AI, and computer-vision systems, with a focus on reliable, production-oriented AI applications.

Technical Skills

Programming: Python, SQL
Machine Learning & Data: Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, CatBoost, SHAP
Generative AI & NLP: RAG, LangChain, LangGraph, Transformers, Sentence-Transformers, Embeddings, Semantic/Vector Search, FAISS
Backend & Databases: FastAPI, REST APIs, PostgreSQL, pgvector, MongoDB
Computer Vision: PyTorch, COLMAP, 3D Gaussian Splatting, Image/Video Processing
Developer Tools: Git, GitHub, Docker, Linux, Jupyter

Experience

AI/ML Engineer Intern — Jun 2026 – Present
Allora Tech — *Tamil Nadu, India*
– Develop conversational AI applications using **Python, FastAPI, LangChain, and LangGraph**.
– Build **RAG pipelines**, retrieval, conversational memory, and workflow orchestration with **PostgreSQL/pgvector**.

Software Engineer Intern — Jul 2026 – Oct 2026
NxtHive — *Tamil Nadu, India*
– Developed backend features for recruitment-management applications using **TypeScript, Fastify, Prisma, and PostgreSQL**.
– Worked on **REST APIs**, database operations, and application workflows.

Artificial Intelligence Engineer Intern — Mar 2026 – Jun 2026
InnoXR Labs Pvt. Ltd. — *Chennai, India*
– Built a **3D reconstruction pipeline** integrating frame extraction, COLMAP, Gaussian Splatting, and GPU-based training with **PyTorch** and computer-vision workflows.

Data Analyst Intern — Jul 2024 – Oct 2024
India Advocacy — *Chennai, India*
– Developed **Java/Selenium** data pipelines to extract, clean, transform, and analyze **1M+ records**.

Projects

Prompt Aegis: AI Agent Security & Governance — FastAPI, LangGraph, PostgreSQL, Redis, React
– Built an AI-agent security gateway covering authentication, permission checks, policy enforcement, **risk scoring**, rate limiting, and audit logging.

SourceUp: Supplier Search & Ranking System — Research Project – Submitted for IEEE
– Built a supplier search and ranking pipeline for **1.27M records** using **SBERT, FAISS, LightGBM LambdaRank, and SHAP** with **23K+ supplier–query pairs**.

FinGuard Pro: Financial Fraud Detection — Published, IEEE ICAAIC 2025
– Developed an ML fraud detection pipeline on **100K transactions**; compared XGBoost, LightGBM, and CatBoost, achieving **94% recall** and **0.9951 ROC-AUC**.

S.A.F.E.: Crowd Anomaly Detection — Design Thinking Project, 2026
– Evaluated crowd anomaly detection on benchmark datasets; MAD achieved **0.8391 ROC-AUC** and reduced false positives by **40–60%**, served via a **FastAPI** backend.

Education

Sathyabama Institute of Science and Technology — Expected 2027
M.Sc. Computer Science — *Chennai, India*

Sathyabama Institute of Science and Technology — 2025
*B.Sc. Computer Science (Artificial Intelligence) – **First Class*** — *Chennai, India*

Awards & Publications

– **Best Industry Innovation Award**, Industry Innovation '26 – national-level hackathon, 63+ teams (Mar 2026)
– **4th Place**, Origin 24-Hour Hackathon – SIMATS Engineering, Chennai (Apr 2026)
– **IEEE ICAAIC 2025:** FinGuard Pro, financial fraud detection research

Additional Experience

Podcast Head, Philosophy and Open Discussion Club (2023–24) | **Sales Associate**, Bata Singapore (2022) | **Security Trooper**, Singapore Army (2020–22)